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Differentiated mTOR but not AMPK signaling after strength vs endurance exercise in training-accustomed individuals

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The influence of adenosine mono phosphate (AMP)-activated protein kinase (AMPK) vs Akt-mammalian target of rapamycin C1 (mTORC1) protein signaling mechanisms on converting differentiated exercise into training specific adaptations is not well-established. To investigate this, human subjects were divided into endurance, strength, and non-exercise control groups. Data were obtained before and during post-exercise recovery from single-bout exercise, conducted with an exercise mode to which the exercise subjects were accustomed through 10 weeks of prior training. Blood and muscle samples were analyzed for plasma substrates and hormones and for muscle markers of AMPK and Akt-mTORC1 protein signaling. Increases in plasma glucose, insulin, growth hormone (GH), and insulin-like growth factor (IGF)-1, and in phosphorylated muscle phospho-Akt substrate (PAS) of 160 kDa, mTOR, 70 kDa ribosomal protein S6 kinase, eukaryotic initiation factor 4E, and glycogen synthase kinase 3α were observed after strength exercise. Increased phosphorylation of AMPK, histone deacetylase5 (HDAC5), cAMP response element-binding protein, and acetyl-CoA carboxylase (ACC) was observed after endurance exercise, but not differently from after strength exercise. No changes in protein phosphorylation were observed in non-exercise controls. Endurance training produced an increase in maximal oxygen uptake and a decrease in submaximal exercise heart rate, while strength training produced increases in muscle cross-sectional area and strength. No changes in basal levels of signaling proteins were observed in response to training. The results support that in training-accustomed individuals, mTORC1 signaling is preferentially activated after hypertrophy-inducing exercise, while AMPK signaling is less specific for differentiated exercise.

Specific exercise training regimes elicit specific skeletal muscle adaptations, with endurance (ET) training (long duration/low intensity) primarily modulating capacity for substrate utilization and with strength training (ST) (short duration/high intensity) primarily modulating contractile protein synthesis and muscle hypertrophy (Hawley, 2002; Coffey & Hawley, 2007; Spiering et al., 2008b).

During EE especially, but also during single-bout strength exercise (SE), the AMP-activated protein kinase (AMPK) functions both as a cellular energy gauge involved in the acute regulation of substrate utilization during EE and as a nodal point in the transcriptional regulation of metabolic genes (Munday et al., 1988; Jessen & Goodyear, 2005; Wojtaszewski et al., 2005; Fujii et al., 2006; McGee & Hargreaves, 2010a). The latter function is mediated by directly targeting transcriptional regulators, such as the cAMP response element-binding protein (CREB) family of transcription factors (Thomson et al., 2008; Spiering et al., 2008b), the peroxisome proliferator-activated receptor-γ coactivator 1α (PGC1α) transcriptional coactivator (Jager et al., 2007), and the class II histone deacetylases transcriptional repressors (McGee & Hargreaves, 2010b).

By analogy, the mTORC1/raptor complex constitutes a key regulator of exercise-induced protein synthesis in response to SE by controlling the rate of protein translation via the 70 kDa ribosomal protein S6 kinase (P70S6K) and the eukaryotic initiation factor (eIF) 4E binding protein (4E-BP1). This complex is sensitive to nutrient availability, growth factors, and cell deformation and have been speculated to be under the upstream influence of Akt kinase (Sandri, 2008; Spiering et al., 2008b). Furthermore, Akt may also target other substrates involved in muscle cell hypertrophy and/or its metabolic response to growth, such as the Akt substrate of 160 kDa (AS160), glycogen synthase kinase 3 (GSK3) isoforms, and forkhead box O family members of transcription factors (Sandri, 2008; Spiering et al., 2008b).

In a much cited study by Atherton et al. (2005), EE and SE was mimicked by low-frequency and high-frequency electrostimulation on rat muscle (Atherton et al., 2005). This study supported the existence of an AMPK-Akt "switch" mechanism, according to which AMPK and Akt-mTORC1 signaling would be selectively activated by EE and SE, respectively (Atherton et al., 2005). Studies in humans have not been able to strongly verify the existence of a similar straightforward AMPK vs Akt-mTORC1 switch mechanism (Coffey et al., 2006; Wilkinson et al., 2008; Camera et al., 2010). However, one of these studies included only well-trained athletes, who may suffer from low sensitivity for exercise-induced signaling (Coffey et al., 2006). A second study applied unilateral exercise in the fed state with opposite legs performing differentiated exercise immediately after one another (Wilkinson et al., 2008), an approach that may have introduced a common systemic response to influence signaling. A third study only followed subjects 1 h into post-exercise recovery (Camera et al., 2010), thus potentially missing important later signaling events. Furthermore, none of these studies controlled for non-exercise-related stressors that may have influenced signaling responses by including a sedentary control group. Added to that, the regulatory role of Akt on mTORC1 has been recently been questioned, wherefore it may be more relevant to refer to AMPK vs mTORC1 signaling (Goodman et al., 2010; Hamilton et al., 2010).

The aim of the current study was therefore to test the effect of differentiated exercise on the AMPK vs mTORC1 signaling in human skeletal muscle using three precautionary measures. First, subjects were training-accustomed prior to commencing an exercise trial to minimize the influence of exercise-unrelated stressors. Second, because no strict common consensus exists on feeding protocols, single-bout exercise trials and the immediate recovery were completed in the fasted state, but with the inclusion of an independent sedentary non-exercise group to control for exercise-unrelated stressors related to dietary condition, repeated biopsy sampling, and diurnal rhythm. Third, an extended time course of post-exercise sampling was implemented.

We hypothesized that mTORC1 signaling would be selectively activated by SE, whereas AMPK signaling would be activated by both types of exercise but to a relatively higher degree after EE compared with SE.

Materials and methods

Subjects

Twenty-four untrained, young, healthy, male subjects volunteered to participate in the study {body mass, 79.1 ± 2.4 kg; height, 182 ± 0.0 m; age, 23.3 ± 0.6 years; [mean ± standard error of the mean (SEM)]}.

The subjects were excluded if they had engaged in organized EE or ST within the last 6 months prior to inclusion in the study. This was assured partly by a VO_2. max test [inclusion criteria for VO_2. peak < 50, mL/(kg body mass (BM)/min] and partly an interview/questionnaire on prior training history. Other exclusion criteria included any history of musculoskeletal injuries and use of prescription medicine.

All participants were informed of the purpose and the risks related to the study and gave written, informed consent to participate. The study was approved by the ethics committee of Region Midtjylland, Denmark (j. no. M-20080177) and conducted in accordance with the Declaration of Helsinki.

Eighteen of the subjects were randomly divided into ET or ST groups ($n = 9$ per group). The other subjects composed a non-exercise control group ($n = 6$).

Experimental design and training protocol

Prior to commencing the training protocol, the subjects were tested for aerobic and strength performance measures and underwent magnetic resonance imaging (MRI) scanning of the thigh muscle. With several days interspacing these procedures, a venous blood sample and a muscle biopsy from the vastus lateralis muscle were also taken by Bergström needle technique under local anesthesia of the skin and fascia. After a few days of recovery, the exercise groups then completed either 10 weeks of ET on a cycle ergometers or 10 weeks of progressive conventional ST for lower extremity muscle groups.

In brief, ET was performed on stationary bicycles (Kettler Ergoracer GT, Kettler, Ense-Parsit, Germany). Based on the pre- and midway determinations of VO_2 max, maximal workload in watts, and maximal heart rate, target watt and target heart rate for each training session were used to aid training intensity. Three weekly sessions were composed of one session each of continuous cycling of 30–45 min at 60–75% of watt max, 1-s session composed of two intervals of 20 min at 70–80% of watt max interspaced by 5 min of light cycling, and a third session composed of 8 × 4 min intervals at 80–90% of watt max interspaced by 1-min light cycling.

The resistance training group completed a conventional progressive overload training program, similar to that previously described, composed of 3 leg exercises each performed as 3–5 sets of 10 → 4–6 repetitions, with repetitions corresponding to repitition maximum (RM) loading (Vissing et al., 2008). The recovery time between sets was set to 2 min during the first 15 training sessions, increasing to 3 min for the remaining 15 training sessions. During the last 15 sessions, the subjects were instructed to perform the concentric part of the exercises as fast as possible.

All training sessions were supervised to ensure proper progression for both groups, and all training sessions were separated by at least one training-free day.

After 3 days of recovery, the training period was followed by completion of a single-bout EE or a single-bout SE session.

Single-bout exercise protocol

The subjects had been instructed to abstain from physical activity for at least 3 days prior to the trial. A schematic overview of trial procedures immediately during pre-exercise, during exercise, and during the 22-h recovery period is shown in Fig. 1.

The subjects arrived after an overnight fast at 08:00 am on day 1 for pre-exercise measurements. The subjects rested in the supine position for approximately 30 min. A venous catheter was then inserted into an antecubital vein and a pre-exercise blood sample (basal) was taken. A pre-exercise muscle biopsy (basal) was then taken from the vastus lateralis muscle.

At 08:30 am, the subjects of the endurance group commenced 120 min of bicycle exercise at 60% post-training Vo_2 peak. The subjects in the strength group remained resting until 10:00 am before commencing 4 × post-training 12 RM of three thigh muscle exercises with 1.5-min rest between sets. The three thigh muscle exercises were leg press, knee extension, and hamstring curl. Minor weight adjustments were allowed to ensure performance of 12 RM as precise as possible. Non-exercise control subjects rested from 08:30 am to 10:30 am. The non-exercise control group was included to control for potential non-exercise-related effects of

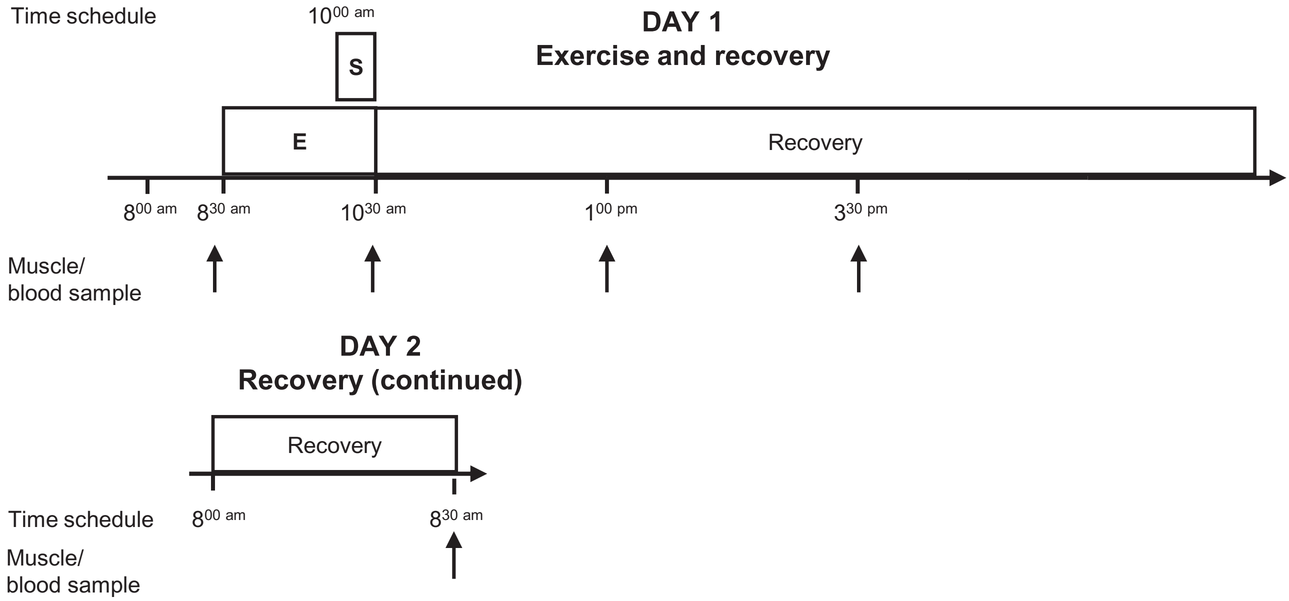

Fig. 1. Single-bout experimental protocol. The subjects arrived after an overnight fast at 08:00 am on day 1 for pre-exercise measurements. At 08:30 am, the subjects of the endurance group (E) commenced 120 min of bicycling exercise at ~60% VO_2 peak. The subjects of the strength group (S) remained resting until 10:00 am before commencing 4 × 12 RM of three thigh muscle exercises with 1.5 min rest between sets. The non-exercise control subjects rested from 8:30 am to 10:30 am. After exercise, all of the subjects rested under fasting conditions until 15:30 am (corresponding to 5 h post-exercise) on day 1 where after meals were allowed until 22:00 am. The subjects arrived after an overnight fast at 08:00 am on day 2 for final measurements. Muscle and blood samples were harvested prior to and at 0, 2.5, 5, and 22 h post-exercise. Biopsies corresponding to pre-exercise and at 2.5 and 22 h after exercise were harvested from one leg, while biopsies corresponding to 0 and 5 h after exercise were harvested from the opposite leg.

fasting conditions and the invasive procedures inherent of the protocol. Furthermore, to control for potential effects of circadian rhythm, it was attempted to strictly equalize absolute daily time points as well as time resolution of the protocol regardless of which group the subjects were assigned to.

After ± exercise, all of the subjects rested under fasting conditions (with water offered *ad libitum*) until 15:30 am (corresponding to 5-h post-exercise) on day 1, where after meals by own choice were allowed until 22:00 am. The subjects arrived after an overnight fast at 8:00 am on day 2 for final measurements. Muscle and venous blood samples were harvested prior to and at 0, 2.5, 5, and 22 h post-exercise.

Biopsies were sampled from separate incision holes. For the biopsy corresponding to the 0 h time point, local anesthesia of the skin and fascia was performed during a 30-s break approximately 10 min before termination of exercise, and the biopsy was taken within seconds after termination of exercise. The biopsy was sampled at 2.5 h and was taken from the same leg as the pre-exercise biopsy, whereas the muscle biopsies sampled at 0 and 5 h post-exercise were taken from the opposite leg. Within each leg, a minimum distance of 3 cm between biopsy sampling sites was ensured, and similar sampling depth of biopsies was attempted.

On day 2, the subjects arrived at the laboratory at 08:00 am after an overnight fast. They again rested for 30 min in the supine position, and final blood sample and muscle biopsy were obtained corresponding to the 22-h post-exercise time point. This muscle biopsy was sampled from the opposite leg compared with the pre-exercise muscle biopsy.

Exercise capacity measurements

Measures related to capacity for aerobic/EE

VO_2 peak was determined before and after training on all subjects through an incremental exercise test on a Monark Ergomedic 834E bicycle ergometer (Monark ergometic 894 E, Monark, Varberg, Sweden), with rates of oxygen uptake and carbon dioxide release determined by an online respiratory gas exchange analyzer (AMIS 2001, Innovision, Odense, Denmark). Watt max was estimated to principles of Andersen (1995). Maximal heart rate and submaximal heart rate at 140 W were monitored continuously with a heart rate monitor (Polar, Oulu, Finland).

Measures related to capability for SE

Maximal dynamic knee extensor strength (determined as 3 RM knee extensor strength), knee extensor muscle cross-sectional area (anatomical cross-sectional area (CSA), determined by MRI), and mean fiber CSA (physiological CSA, determined by ATPase histochemistry) were determined as previously described (Vissing et al., 2008). Peak torque was determined in an isokinetic dynamometer (Humac Norm, CSMI, Stoughton, Wisconsin, USA) with 90-degree hip flexion. The protocol was composed of three maximal contractions at three different angular velocities: 30, 90, and 180 degrees/s, with standardized verbal encouragement given during the contractions. All contractions were interspaced with 1 min of recovery time. All trials were sampled at 100 Hz. Peak torque from the best of the three trials was used for further analysis.

Whole-muscle cross-sectional area analysis – MRI

All imaging was performed with a 1.5-T scanner (Philips Achieva, Best, the Netherlands). The subjects were placed in supine position and the MRI scans were performed on the left leg using a cardiac coil. A T1-weighted, fast spin echo sequence was conducted with use of the following parameters: scan matrix = 288 × 282, field of view = 230 × 230 mm^2, number of slices = 20, slice thickness = 7.5 mm, slice gap = 1 mm, repetition time = 2 s, echo time = 5.3 ms, echo train length = 18, number of signal averages = 3, and scan time = 3 : 12 min. After an initial frontal scout scan, 20 transversal slices were acquired, with a slice at a position corresponding to one half of the femur length chosen for CSA determination. CSA of the knee extensor compartment (mm vastus lateralis, vastus medialis, vastus intermedius, and rectus femoris) was calculated using custom-made software. The analysis program was validated by comparing area analyses on phantom scans with analyses made with the standard Philips radiological analysis environment (ViewForum rel. 5.1, 2006, Philips, Best, Netherlands).

Muscle fiber analysis – ATPase histochemistry

Biopsies were obtained from the middle section of the vastus lateralis muscle by applying the Bergström needle technique. The muscle samples were dissected free of visible fat and connective tissue, and a part of the biopsy was immediately mounted with Tissue-Tek (QIAGEN, Valencia, California, USA), frozen in isopentane cooled with liquid nitrogen and stored at −80°C until further analysis. Serial sections (10 µm) were cryocut, and ATPase histochemistry analysis was performed as described previously (Brooke & Kaiser, 1970) to enable determination of fiber CSA using a Leica DM2000 microscope (Leica, Stockholm, Sweden) and a Leica high-resolution Color digital firewire camera camera (Leica) combined with image analysis software (Leica Qwin ver. 3, Leica) for visualization and analysis as described by Dalgas et al. (2010).

Determination of plasma substrates and hormones

Serum samples were frozen and stored at −20°C. Glucose was measured in duplicate on a Beckman Glucoanalyzer (Beckman Instruments, Palo Alto, California, USA); GH, IGF-I, and insulin were analyzed using time-resolved fluoroimmunoassay (TR-IFMA; AutoDELFIA, PerkinElmer, Turku, Finland). Free fatty acids (FFAs) were analyzed by a commercial kit (Wako Chemicals, Neuss, Germany). Analysis was performed corresponding to time point prior to and at 0 and 5 h post-exercise.

Preparation of muscle biopsies

Muscle biopsies were quickly divided into smaller parts immediately after sampling. Pre- and post-training biopsies were divided in one larger part that was immediately frozen in liquid nitrogen, and another part that was immediately mounted with Tissue-Tek and frozen in isopentane cooled with liquid nitrogen. Biopsy samples harvested during the single-bout exercise trial were frozen in liquid nitrogen as quickly as possible (especially at 0 h post-exercise). All samples were then stored at −80°C until further investigation.

Frozen muscle biopsies (50 mg) were homogenized in ice-cold solubilization buffer containing 20 mM Tris, 50 mM NaCl, 5 mM $Na_4P_2O_7$, 50 mM NaF, 250 mM sucrose, 2 mM DTT, 1% Triton-X 100, 2 µg/mL aprotinin, 5 µg/mL leupeptin, 0.5 µg/mL pepstatin, 10 µg/mL antipain, 1.5 mg/mL benzamidine, and 100 µmol/L 4-(-2-aminoethyl)-benzenesulfonyl fluoride, hydrochloride (pH 7.4). Insoluble materials were removed by centrifugation at 16 000 × g for 20 min at 4 C and protein content on the supernatant was determined.

Immunoblotting

Equal amounts of protein were resolved by Sodium Dodecyl Sulphate Poly-Acrylamide Gel Electrophoresis using the Bio-Rad Criterion blotting system and Bio-Rad precast gels (Criterion XT 4–15% Bis-Tris Gels, Bio-Rad, Hercules, California, USA). All time points from each of two subjects (subjects randomized between gels) were run on the same gel. Proteins were transferred to Polyvinylidene fluoride membranes (Bio-Rad), blocked in either 2% fat-free milk (SIGMA-ALDRICH, St. Louis, Missouri, USA) or 1% bovine serum albumin in 1 × tris buffered saline with tween, and incubated with primary antibody. The following were purchased from Cell Signaling (Beverly, California, USA): Akt, phosphospecific Akt (Ser^{473} and Thr^{308}); GSK3 (α and β), phosphospecific GSK3 (α and β); mTOR, phosphospecific mTOR (Ser^{2448}); P70S6k, phosphospecific P70S6k (Thr^{389}); eIF4E, phosphospecific eIF4E (Ser^{209}); PAS; AS160; CREB, phosphospecific CREB (Ser^{133}); ACC, phosphospecific ACC (Ser^{79}); AMPK, phosphospecific AMPK (Thr^{172}); and HDAC4/5, phosphospecific HDAC5 antibodies. Phosphospecific HDAC5 antibodies (Ser^{259}) have been described previously (Munday et al., 1988). Horseradish peroxidase-conjugated goat anti-rabbit IgG antibody (Pierce Chemical, Rockford, Illinois, USA) was used as secondary antibody. Proteins were visualized by BioWest-enhanced chemiluminescence (Pierce Supersignal, Thermo Fisher Scientific, Rockford, Illinois, USA) and quantified using UVP BioImaging System (UVP, Upland, California, USA). Analysis was performed corresponding to time-points prior to and at 0, 2.5, 5, and 22 h post-exercise. Phosphoproteins were normalized with total protein for the same blot (probing, stripping, and reprobing) for each target. Pre ± exercise values were set to a fixed value of 100%, and post-exercise samples were normalized to this fixed value. Examples of Western blots are shown in Fig. 2.

Statistical analyses

Data are presented as mean ± SEM. All statistical analyses were performed using SigmaPlot (SigmaPlot v 11.0, Systat Software Inc., Chicago, Illinois, USA). For variables independent of time, Student's *t*-test was performed to test for differences in basal levels between groups. For variables measured before and after the training period and during single-bout exercise trials, two-way analysis of variance with repeated measures was performed to test for changes due to group, time, or group × time. When a significant effect of group and time was found, significant pairwise differences were detected using Student Newman Keul's post-hoc test. Protein signaling data are presented as arbitrary units. For all statistical tests, $P < 0.05$ was considered significant.

Results

Two subjects from each training group did not complete the training period because of exercise-unrelated circumstances, leaving $n = 7$ in each training group for all data presented.

Measures related to exercise capacity

Changes in exercise capacity parameters as induced by 10 weeks of training are shown as absolute values ± SEM in Table 1, while only relative changes are reported below. No differences were observed between groups prior to commencing the training period for any of these variables.

In measures of exercise capacity, the ET group exhibited training-induced increases in VO_2 peak (~11%; $P < 0.01$) and in maximal workload (~13%; $P < 0.001$), and a training-induced decrease in submaximal intensity exercise heart rate (~13%; $P < 0.01$). No changes were observed in response to ST. Other than a tendency to a group difference for submaximal intensity exercise heart

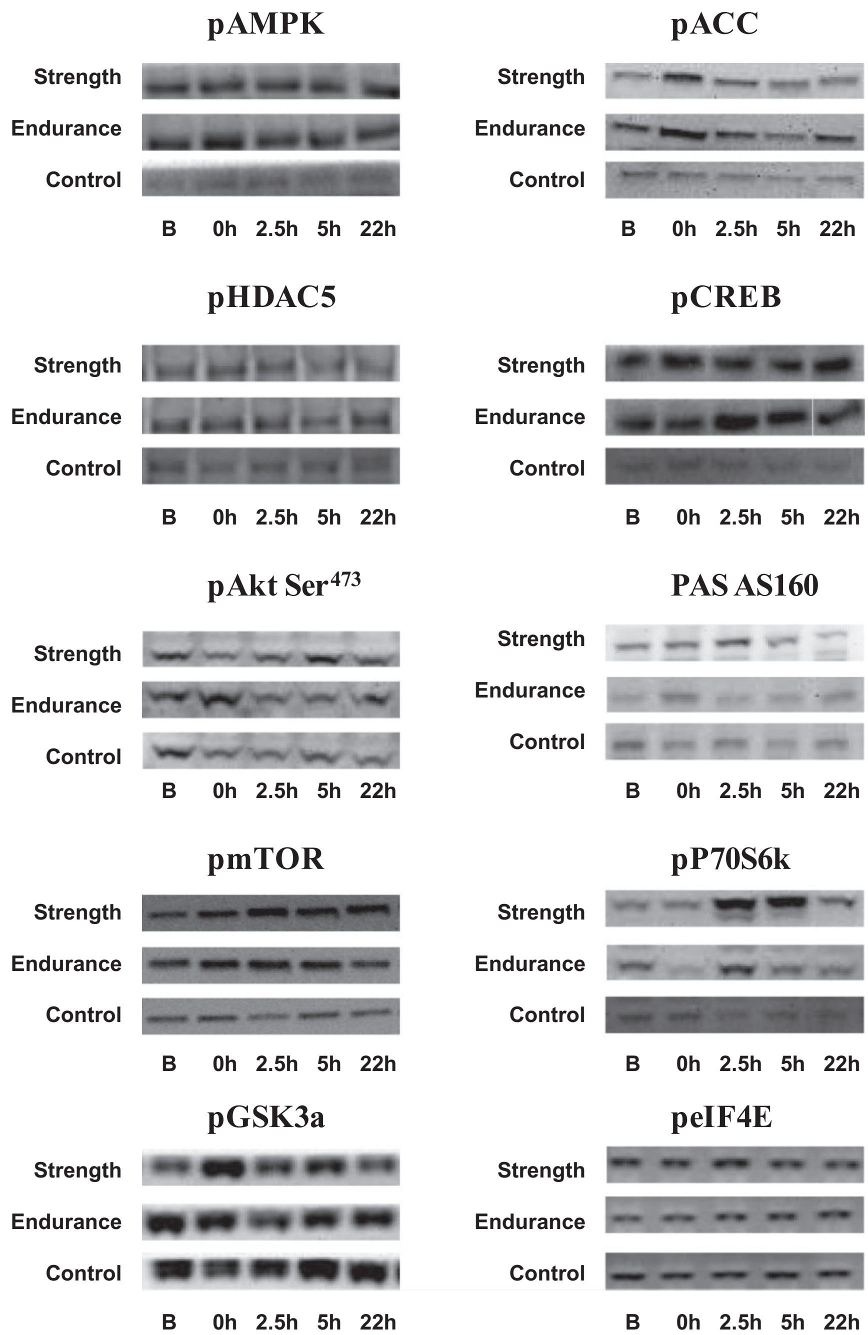

Fig. 2. Examples of Western blots. Examples of representative Western blots are shown for time course effects on Akt-mTOR signaling and AMP-activated protein kinase signaling proteins in response to strength, endurance, and control interventions, respectively.

rate (i.e. ET < ST) ($P = 0.098$), no changes between groups were observed.

The ST group exhibited training-induced increases in knee extensor muscle CSA (~9%; $P < 0.01$), in fiber CSA (~18%; $P < 0.05$), in peak torque (~19%; $P < 0.001$), and in maximal knee extensor strength (~39%; $P < 0.001$). No changes were observed in response to ET. Changes between groups (i.e. ST > ET) were observed for peak torque ($P < 0.05$) and maximal knee extensor strength ($P < 0.01$).

Plasma substrates and hormones

Acute responses in plasma substrates and hormones from the ± single-bout exercise intervention are shown as absolute values ± SEM in Table 2, while only relative changes are reported below.

A group × time interaction was observed for plasma FFA ($P < 0.001$). In response to EE, plasma FFA increased by ~409% ($P < 0.001$) from basal levels to 0 h after exercise, then decreasing to ~160% above basal

Table 1. Measures related to exercise capacity

	Ten weeks endurance training		Ten weeks strength training	
	Pre	Post	Pre	Post
Measures of capacity for aerobic/endurance exercise				
$VO_{2\ peak}$, mL (kg BM)/min	44.9 ± 1.4	49.7 ± 2.0**	46.9 ± 3.0	46.9 ± 2.6
Maximal workload, watt	330 ± 18	374 ± 18***	322 ± 20	332 ± 19
Heart rate, bpm at 140 W	155 ± 3	138 ± 3**(#)	162 ± 9	152 ± 7
Measures of capacity for strength exercise				
Knee extensor muscle CSA, cm^2	82 ± 4	85 ± 4	77 ± 3	84 ± 2**
Fiber CSA, μm^2	5446 ± 391	5781 ± 232	5207 ± 266	6125 ± 342*
Peak torque at 30°/s, Nm	202 ± 20	194 ± 18	214 ± 13	255 ± 13***#
3 RM knee extension, kg	93 ± 9	94 ± 8	96 ± 5	133 ± 4***##

Data are shown as mean ± standard error of mean.
*Training effect within group, $P < 0.05$, **Training effect within group, $P < 0.01$, ***Training effect within group, $P < 0.001$.
(#)Tendency to a difference between groups, $P < 0.1$, #Difference between groups, $P < 0.05$, ##Difference between groups, $P < 0.01$.
CSA, cross-sectional area; RM, repetition maximum.

Table 2. Plasma substrates and hormones

	Endurance group			Strength group			Control group		
	Pre	0 h	5 h	Pre	Post	5 h	Pre	0 h	5 h
Free fatty acid (mmol/L)	0.4 ± 0.04	1.5 ± 0.2E*	1.0 ± 0.2E	0.5 ± 0.1	0.5 ± 0.1	1.0 ± 0.1S	0.4 ± 0.1	0.5 ± 0.1	0.8 ± 0.1C
Glucose (mmol/L)	4.4 ± 0.1	4.7 ± 0.2	4.2 ± 0.2	4.7 ± 0.1	6.2 ± 0.2S*	4.8 ± 0.1	4.4 ± 0.2	4.6 ± 0.2	4.4 ± 0.2
Insulin (pmol/L)	32.9 ± 5.2	52.7 ± 14.9	20.2 ± 4.6	34.0 ± 2.8	88.8 ± 14.6S*	25.1 ± 3.5	31.6 ± 6.9	30.7 ± 5.8	25.5 ± 7.4
Growth hormone (ng/mL)	0.1 ± 0.1	3.2 ± 0.9†	0.9 ± 0.3	2.0 ± 1.7	10.5 ± 2.6S*	0.5 ± 0.1	0.4 ± 0.2	0.1 ± 0.1	0.2 ± 0.1
Insulin-like growth factor-I (μg/L)	254 ± 18	267 ± 21	253 ± 19	300 ± 19	330 ± 22S	307 ± 22	261 ± 19	280 ± 21	273 ± 22

Plasma concentrations are shown ± standard error of the mean, prior to, and at 0 and 5 h after endurance exercise, strength exercise, or non-exercise control intervention. E, S, and C denote a difference from basal level within the endurance, strength, and control groups, respectively, $P < 0.05$.
*Denotes a difference from all other groups, $P < 0.05$.
†Denotes a tentative difference from control group, $P < 0.05$.

level at 5 h after exercise ($P < 0.001$). No changes were observed in response to SE or non-exercise control interventions at 0 h, but both groups increased (by ~80–105%, both $P < 0.001$) at 5 h after intervention. At 0 h, SE was significantly increased compared with two other groups ($P < 0.01$).

Group × time interactions were observed for plasma glucose, plasma insulin, and plasma GH (all $P < 0.001$). In response to SE, plasma glucose increased by ~32% ($P < 0.001$), plasma insulin increased by ~161% ($P < 0.001$), and plasma GH increased by ~514% ($P < 0.001$) at 0 h after exercise. No changes were observed in response to EE or non-exercise control interventions. Group differences were observed at 0 h between SE and the two other groups ($P < 0.001$).

An overall time effect was observed for plasma IGF-1 ($P < 0.001$). In response to SE, plasma IGF-1 increased by ~10% ($P < 0.01$) from basal levels to 0 h after exercise. No changes were observed in response to EE or non-exercise control interventions. No group differences were observed at any time point.

AMPK signaling

Changes in phosphorylation of AMPK signaling proteins in response to single-bout exercise are shown as representative blots in Fig. 2 and as graphs in Fig. 3.

A group × time interaction was observed for AMPK phosphorylation ($P < 0.05$). In response to EE and SE, pAMPK increased by ~44% ($P < 0.001$) and ~10% ($P < 0.05$), respectively, from basal levels to 0 h after exercise, then returning to basal levels throughout the remainder of the recovery time course. A difference was observed between EE and non-exercise control intervention at 0 h ($P < 0.05$). No other differences within or between groups were observed (Fig. 3(a)).

A group × time interaction was observed for ACC phosphorylation ($P < 0.05$). In response to EE, pACC increased by ~123% ($P < 0.001$) from basal levels to 0 h after exercise, then returning to basal levels throughout the remainder of the recovery time course. A difference was observed between both EE and SE to non-exercise control intervention at 0 h ($P < 0.001$ and $P < 0.01$,

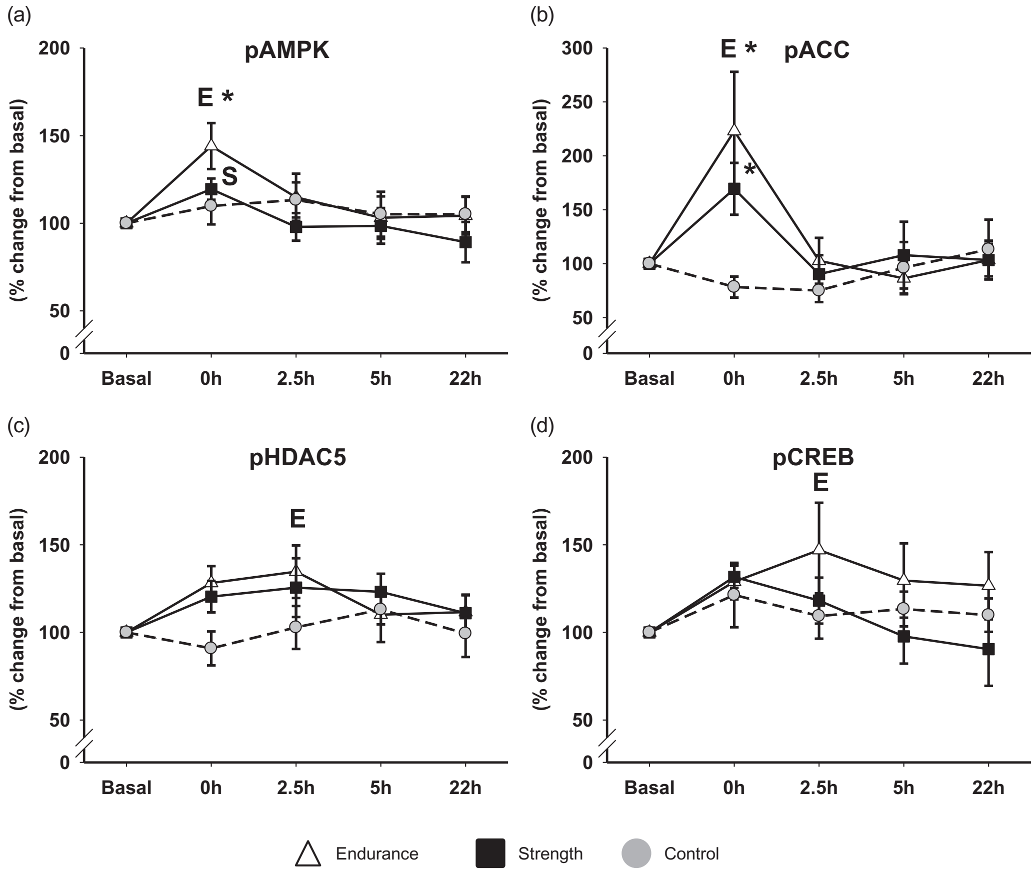

Fig. 3. AMP-activated protein kinase (AMPK) signaling. Protein phosphorylation results are shown as % change ± standard error of the mean from basal set to a fixed value of 100%, at 0, 2.5, 5, and 22 h post-exercise recovery period after endurance exercise, strength exercise, or non-exercise control intervention for (a) 5′ pAMPK, (b) cAMP response element-binding protein (pCREB), (c) histone deacetylase 5 (pHDAC5), and (d) acetyl-CoA carboxylase (pACC). E denotes differences from basal level within the endurance group, $P < 0.05$. S denotes differences from basal level within the strength group, $P < 0.05$. * denotes difference from control group, $P < 0.05$.

respectively). No other differences within or between groups were observed (Fig. 3(b)).

Overall time effects were observed for phosphorylated HDAC5 and CREB ($P < 0.001$ and $P < 0.05$, respectively). In response to EE, pHDAC5 increased by ~28% ($P < 0.05$) and pCREB increased by ~47% ($P < 0.05$) at 2.5 h after exercise before returning to basal levels throughout the remainder of the recovery time course. No changes were observed in response to SE or non-exercise control interventions. No other differences within or between groups were observed (Fig 3(c) and (d)).

Ten weeks of training did not alter protein expression of any of the reported signaling proteins (data not shown).

Akt-mTORC1 signaling

Changes in phosphorylation of Akt-mTORC1 signaling proteins in response to single-bout exercise are shown as representative blots in Fig. 2 and as graphs in Fig. 4.

A group × time interaction was observed for $pSer^{473}$ Akt ($P < 0.01$). In response to EE, a trend toward a minor increase in $pSer^{473}$ Akt was observed at 0 h after exercise, which was not different from basal levels, yet different from all other groups at 0 h ($P < 0.05$), where a trend toward a decrease was observed. $pSer^{473}$ Akt for EE then decreased to ~62% of basal levels at 5 h after exercise ($P < 0.05$), while SE and non-exercise control groups did not differ from basal levels this late in recovery (Fig. 4(a)). Akt^{Thr308} did not change with either intervention (results not shown).

Group × time interactions were observed for AS160 PAS signal, mTOR phosphorylation, and P70S6K phosphorylation ($P < 0.01$, $P < 0.05$, and $P < 0.05$, respectively). In response to SE, the AS160 PAS signal increased by ~114% ($P < 0.01$), pmTOR increased by ~91% ($P < 0.01$), and P70S6K phosphorylation increased by ~281% ($P < 0.001$) during the time course from 2.5 to 22 h during recovery. No other differences within or between groups were observed (Fig 4(b), (c), and (d)).

Overall time effects were observed for GSK3α phosphorylation and eIF4E phosphorylation (both $P < 0.001$). In response to SE, pGSK3α increased by ~20% ($P < 0.05$) at 0 h after exercise eIF4E increased by ~25% ($P < 0.05$) at 5 h after exercise. No other

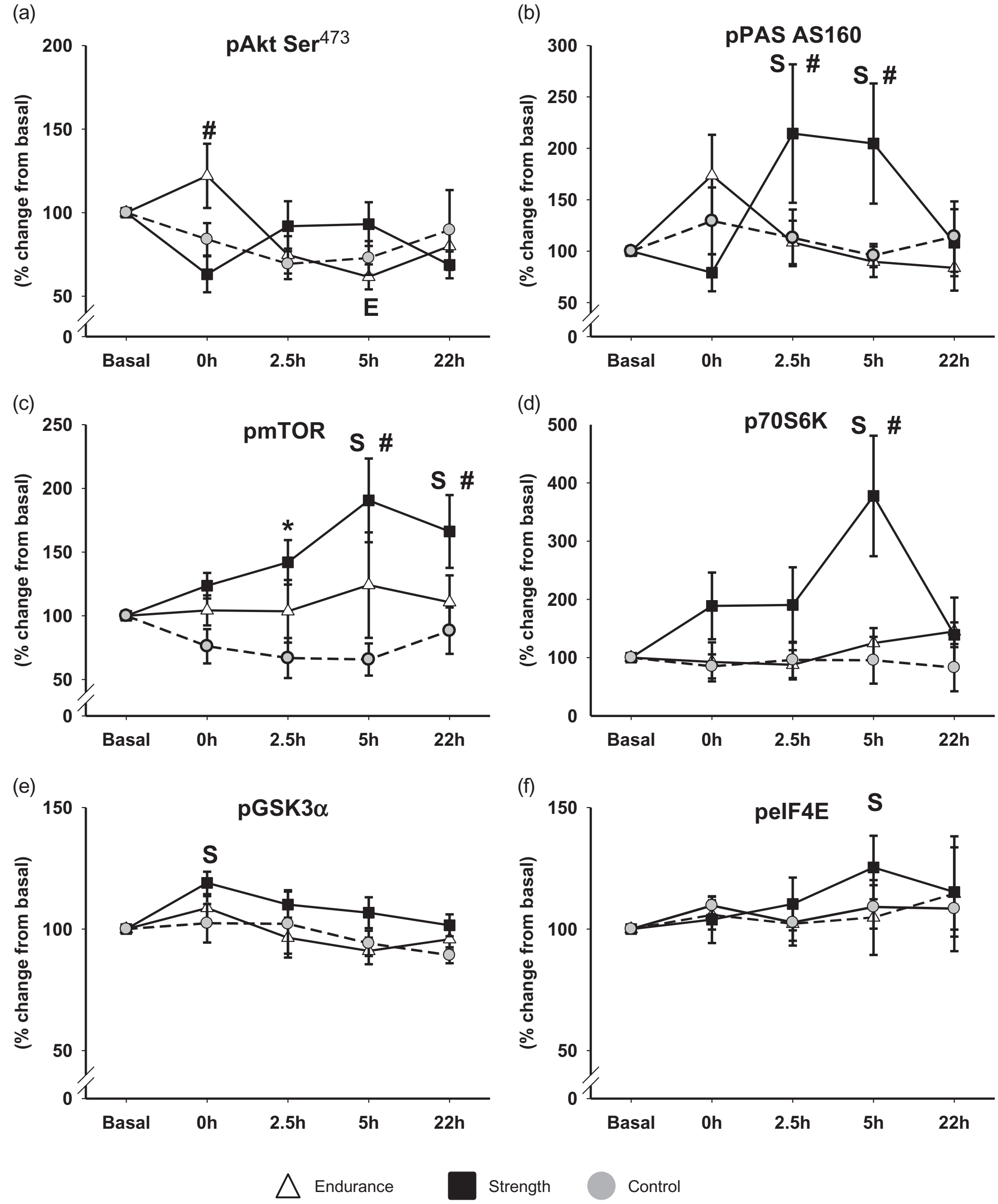

Fig. 4. Akt-mTOR signaling. Protein phosphorylation results are shown as % change ± standard error of the mean from basal set to a fixed value of 100%, at 0, 2.5, 5, and 22 h post-exercise recovery period after endurance exercise, strength exercise, or non-exercise control intervention for (a) Akt kinase (pAkt), (b) Akt substrate of 160 kDa (PAS AS160), (c) mammalian target of rapamycin (pmTOR), (d) P70S6 kinase (P70S6K), (e) eukaryotic translation initiation factor 4E (peIF4E), and (f) glycogen synthase kinase 3α (pGSK3α). S denotes differences from basal level within the strength group, $P < 0.05$. E denotes differences from basal level within the endurance group, $P < 0.05$. # denotes difference from all other groups, $P < 0.05$. * denotes difference from control group, $P < 0.05$.

differences within or between groups were observed (Fig 4(e) and (f)). pGSKβ did not change with any intervention (results not shown).

Ten weeks of training did not alter protein expression of any of the reported signaling proteins (data not shown).

Discussion

The main finding of the present study was that in training-accustomed individuals, activation of several components of mTORC1 and downstream signaling was activated exclusively from SE, whereas AMPK signaling responded less specific to differentiated exercise. The activation of components of the mTORC1 pathway was demonstrated by appliance of a more elaborate time course than what has previously been applied in human studies on AMPK vs Akt-mTOR signaling, and because we included a non-exercise control group, we provide strength to the conclusion that these signaling responses were in fact due to the differentiated exercise stimulation *per se* and not due to exercise-unrelated stimuli inherent of the study design.

In relation to the acute Akt-mTORC1-signaling responses, except for Akt itself, our results displayed a very distinctive pattern. Accordingly, phosphorylation of PAS AS160, mTOR, and P70S6K all increased exclusively in response to SE, and with no within-group effects emerging from neither EE nor control intervention. Added to that, phosphorylation of GSK3α and eIF4E exhibited within-group increases in response to SE. Akt^{Ser473} was not observed to increase in response to SE, supporting previous observations after SE in the fasted state (Deshmukh et al., 2006) as well as supporting recent evidence that downstream mTORC1 hypertrophy signaling may not be dependent on a growth factor PI3K-Akt axis, but rather rely on an interplay with mechanotransducers (Goodman et al., 2010; Hamilton et al., 2010). PAS AS160 is a known target of Akt as well as multiple other kinases (Kramer et al., 2006) so the fact that PAS AS160 was significantly upregulated in response to SE in later recovery suggests that upstream kinases other than Akt were responsible for this increased level of phosphorylation. As for GSK3α, its phosphorylation observed immediately after SE is in accordance with an increase in protein synthesis (Spiering et al., 2008b), but from our results, its expected reliance on Akt relative to other upstream kinases seems equally questionable as for PAS AS160. As for the phosphorylation of eIF4E observed at 5 h after SE, this response is logical in timing and to be expected to promote protein synthesis (Spiering et al., 2008b). The response of eIF4E was modest compared with mTORC1 and P70S6K but is in line with observations of elongation being more responsive to amino acid availability than to contractile activity during SE, whereas it is not the case for mTORC1-signaling upstream from elongation (Holm et al., 2010).

In a comprehensive study by Wilkinson et al. (2008), the effects of divergent exercise type and mode on signaling were addressed in a protein-fed state with different legs of the subjects performing differentiated exercise one leg immediately after the other. No major differences in signaling responses emerged from this approach. However, the marked and independent effect of branched chain amino acid containing protein supplementation on mTOR signaling may likely have influenced any exercise-dependent differences (Blomstrand et al., 2006; Rennie et al., 2006), but neither this nor the potential humoral effects of consecutive leg exercise were controlled for by the authors. Only one other human study that we are aware of has investigated signaling responses to differentiated exercise in independent exercise groups in the fasted state. In this recent study, Akt-mTORC1 signaling was quite equally provoked by EE and SE (Camera et al., 2010). However, this study only followed the very early responses (i.e. a detailed time course during the first 1 h into post-exercise recovery) and included subjects unaccustomed to the exercise protocol (Camera et al., 2010), whereas our own results seem to justify the importance of following mTORC1 signaling during an extended post-exercise recovery period and/or inclusion of training-accustomed individuals. Interestingly, one recent study also observed increases in protein synthesis and hypertrophy signaling after EE (Mascher et al., 2011). This study encompassed a higher relative EE intensity compared with our protocol, and also, the subjects were not accustomed to the exercise stimuli through prior training. Therefore, our approach and our results strengthen the contention that several components of mTORC1 and downstream signaling are seemingly activated exclusively by SE in the fasted state when individuals are accustomed through prior training. However, it needs to be stressed that our protocol took the approach of conducting exercise training as it is commonly practiced if improvements in either muscle hypertrophy or muscle endurance capacity are to be obtained. Other less traditional training strategies (i.e. including other choices of intensity, duration, volume, interspaced recovery during exercise, etc.) are likely able to produce responses and adaptations elsewhere in between opposite ends of a continuum (Hulmi et al., 2012; Mascher et al., 2011).

In relation to the acute AMPK signaling responses, EE activated AMPK as well as AMPK substrates, ACC, CREB, and HDAC5, thus facilitating acute regulation of substrate metabolism (ACC) as well as transcriptional regulation of metabolic genes (HDAC5 and CREB). SE was only observed to produce a minor increase in pAMPK and no changes in phosphorylation of ACC, CREB, and HDAC5. On the other hand, for none of the AMPK signaling proteins, were phosphorylation levels observed to be statistically different from those observed in response to EE. Thus, a relative degree of homogeneity exists in activation of AMPK as well as the AMPK substrates inherent of our investigation. While similar results for AMPK have been demonstrated earlier in response to EE (Egan et al., 2010), the acute effects of SE on AMPK signaling are less clear, and to our knowledge, no studies have investigated the effects of SE on activation of CREB and HDAC5 in human skeletal muscle. In conclusion, both EE and SE seem able to switch on AMPK signaling, which seem logically explained by the increase in ATP turnover inflicted from both types of exercise. AMPK activation is proposed to phosphorylate raptor, a component of the mTOR complex suggested to simultaneously switch off mTORC1 activity (Gwinn et al., 2008). AMPK activation induced by EE may this way serve to reduce energy consumption related to protein synthesis under circumstances of combined energy-consuming exercise and fasting. Analysis of raptor was not included in the present study, but because we did not observe an outright downregulation in phosphorylation levels for mTORC1 and downstream signaling after EE, our data do not immediately support such a responsible role of AMPK in switching off hypertrophy signaling under energy-demanding circumstances. On the other hand, SE-induced mTOR signaling was not observed to be switched on until AMPK signaling had subsided so this potential regulatory role of AMPK cannot be ultimately concluded from our results.

SE is known to provoke transient increases in circulating growth hormones, then proposed to activate PI-3 K and hypertrophy signaling (Spiering et al., 2008a, b). Our data immediately seem to support a causal relationship between SE-elicited increases in plasma growth hormones. However, such a causality has been challenged from studies on the effect of high hormone (HH) vs low hormone (LH) trials (i.e. high vs low muscle volume recruited during SE training), in which increased levels of circulating anabolic hormones emerged without concurrent increases in neither mTORC1-related signaling nor muscle hypertrophy after HH compared with LH (Spiering et al., 2008a; West et al., 2009, 2010). In the studies by West et al. (2009, 2010), subjects were given whey protein supplements (the independent effect of which was not controlled for) and in this regard, it is interesting to note that under fasting condition, long-term HH training has very recently been observed to be superior to LH training in increasing muscle cross-sectional area (Ronnestad et al., 2011). Thus, the possible relationship between SE-elicited increases in plasma growth hormones and muscle hypertrophy seems to call for additional investigation. Yet, while we contend that our study design contributes with valuable information on this aspect, we must concede that our own data do not firmly allow for the hypothesis of a singular direct causal relationship between plasma growth hormones and hypertrophy signaling. This and the fact that we did not observe Akt^{Ser473} phosphorylation to increase with SE rather provide further support to the opinion that PI-3K/Akt-independent mechanisms serve as important driving factors for mTORC1 signaling (Hornberger et al., 2004; Goodman et al., 2010).

A multitude of different exercise-unrelated stressors such as circadian rhythm, dietary conditions, repeat biopsies, emotional factors, environmental factors, and/or influence of immune system factors may potentially influence humoral responses and myocellular regulatory systems (Vissing et al., 2005). The inclusion of a non-exercise control group in the current study serves to strengthen reliability in the observed signaling and endocrine responses. In this regard, one should recall that our protocol implied post-exercise fasting, but also note that our control intervention indicated that no fasting-induced changes, other than an increase in plasma FFA, occurred. Another important part of our protocol implied exercise familiarization. A complete lack of familiarization will expectedly increase the risk of influence from exercise-unrelated stressors. On the other hand, sensitivity for signaling responses might be markedly attenuated by very long-term training (i.e. years of training as in athletes) (Coffey et al., 2006). Our compromise, composed of completion of 10 weeks of training prior to conducting the single-bout trials, allowed us not only to accustom subjects to the exercise stimuli in question but also to verify that both the ET and the ST regimens were competent in producing the expected specific adaptations in exercise capacity parameters (Aagaard & Bangsbo, 2005; McArdle et al., 2007). Still, training was not observed to produce changes in the basal levels in expression or phosphorylation of the investigated signaling proteins. This is partly in opposition with some previous training studies in which altered basal levels of some phosphoproteins has been observed (Leger et al., 2006; Hulmi et al., 2009). In one of these studies, changes primarily occurred after combined ST and whey protein supplementation (Hulmi et al., 2009), and in another, it is not immediately clear how dietary conditions were standardized prior to biopsy sampling (Leger et al., 2006) so further tightly controlled studies on these specific aspects are warranted.

In summary, our results support the notion that traditional SE, but not traditional EE, stimulates acute mTORC1 signaling and suggest that repeated downstream mTORC1 signaling participates in ST-induced muscle hypertrophy. Contrary to this, less specificity exists in acute AMPK signaling for differentiated exercise.

Perspectives

In rats, AMPK vs Akt-mTOR signaling divergence has been suggested to explain conversion of EE stimulation into mitochondrial biogenesis and conversion of SE stimulation into muscle hypertrophy, respectively (Atherton et al., 2005). Similar divergence is not confirmed from human studies (Coffey et al., 2006; Wilkinson et al., 2008; Camera et al., 2010). This may adhere to differences between exogenous electrical stimulation compared with genuine exercise stimulation and/or human individuals being less homogenous compared with the inbreed rodents. However, it may also adhere to aspects of study design in human studies. Accordingly, it is important to rule out potential influences of exercise-unrelated metabolic and traumatic triggers on signaling mechanisms in question. We suggest to diminish the potential influence on such stressors by applying independent groups of training-accustomed individuals and to deduce such potential influence from inclusion of non-intervention controls. From our approach, our results position us between the contention of outright exercise-specific AMPK and mTORC1 signaling divergence and the contention of virtually no divergence. Rather, we support the contention that in humans, mTORC1 signaling is induced by hypertrophy-stimulating exercise and/or high-intensity exercise but not low-intensity EE, whereas divergent exercise types are more equally capable of provoking AMPK signaling.

Key words: resistance exercise, AMPK substrates, acute response, hypertrophy signaling, protein synthesis.

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